Eigenvalues and diagonalization

Linear Algebra · Lecture 7

Last time (Lecture 6): orthogonality. Results of Lectures 1–5 are cited where they are used; F1–F4: facts about polynomials, without proof (Chapter 7 of the textbook). $I = I_n$, $e_j = \mathrm{col}_j(I)$.

Polynomials. $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$, $i^2 = -1$; $zw = 0 \Rightarrow z = 0$ or $w = 0$. $f = \sum_k c_k t^k$, $g = \sum_k d_k t^k$ ($c_k, d_k \in \mathbb{C}$, $c_k = d_k = 0$ for $k > m$; f real if all $c_k \in \mathbb{R}$): $f = g$ if all $c_k = d_k$; $f + g$, fg have the coefficients $c_k + d_k$, $\sum_{i+j=k} c_i d_j$; constant term c_0; $\deg f = \max\{k : c_k \neq 0\}$ ($f \neq 0$), monic if $c_{\deg f} = 1$. Values $f(z) = \sum_k c_k z^k$ ($z \in \mathbb{C}$), $f(A) = \sum_k c_k A^k$ (f real, $A^0 = I$, $A^{k+1} = A^k A$); z is a root if $f(z) = 0$.

F1. $f, g \neq 0 \Rightarrow fg \neq 0$ and $\deg(fg) = \deg f + \deg g$.
F2. $f \neq 0$ of degree m has at most m roots; so $f(z) = g(z)$ for infinitely many $z \Rightarrow f = g$.
F3. A monic f of degree $n \geqslant 1$ is $(t-\lambda_1)\cdots(t-\lambda_n)$, $\lambda_i \in \mathbb{C}$ unique up to order (fundamental theorem of algebra): roots $\lambda_1, \ldots, \lambda_n$; coefficient of t^{n-1}: $-(\lambda_1 + \cdots + \lambda_n)$; constant term $(-1)^n \lambda_1 \cdots \lambda_n$.
F4. $(f + g)(z) = f(z) + g(z)$, $(fg)(z) = f(z)g(z)$ ($z \in \mathbb{C}$).

1. Eigenvalues and eigenvectors

Definition 1. Let $A \in \mathbb{R}^{n\times n}$. $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if $Ax = \lambda x$ for some $x \in \mathbb{R}^n$, $x \neq 0$; such x is an **eigenvector** belonging to λ. The **eigenspace** of λ is the subspace $\mathcal{N}(A - \lambda I) = \{x \in \mathbb{R}^n : Ax = \lambda x\}$ (Lecture 4, Definition 7: null spaces are subspaces); its nonzero vectors are the eigenvectors belonging to λ.

Eigenvalues need not be real numbers: once the characteristic polynomial of A is defined below, the eigenvalues of A are defined as the n complex roots of that polynomial, listed with multiplicity, and the eigenvalues of this definition are exactly the real ones among them.

Theorem 2. Let $A \in \mathbb{R}^{n\times n}$, $\lambda \in \mathbb{R}$. Then λ is an eigenvalue of $A \iff \det(\lambda I - A) = 0 \iff \mathrm{rank}(A - \lambda I) < n$.

Proof.

The chain of equivalences; $\det(-F) = (-1)^n \det F$ (Lecture 5, Theorem 12(c)):

λ is an eigenvalue	$\iff$ $(A - \lambda I)x = 0$ for some $x \neq 0$	(Lecture 1, Theorem 3(a))
	$\iff$ $A - \lambda I$ is not invertible	(Lecture 2, Theorem 15)
	$\iff$ $\mathrm{rank}(A - \lambda I) < n$	(Lecture 4, Theorem 12(d))
	$\iff$ $\det(A - \lambda I) = 0$	(Lecture 5, Theorem 12(a))
	$\iff$ $\det(\lambda I - A) = 0$	(Lecture 5, Theorem 12(c))

□

2. The characteristic polynomial

Characteristic polynomial. For $A \in \mathbb{R}^{n\times n}$, $\det(tI - A)$, computed by cofactor expansion (Lecture 5, Definition 7) with t as a number, is a real polynomial: the **characteristic polynomial** $p_A(t) = \det(tI - A)$ of A. So $p_A(c) = \det(cI - A)$ for $c \in \mathbb{R}$.

Theorem 3. Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ and $\operatorname{tr} A = a_{11} + \cdots + a_{nn}$.

(a) p_A is monic of degree n; its coefficient of t^{n-1} is $-\operatorname{tr} A$; its constant term is $(-1)^n \det A$.

(b) $\lambda \in \mathbb{R}$ is an eigenvalue of A (Definition 1) $\iff$ $p_A(\lambda) = 0$. So A has at most n such eigenvalues.

Proof. (a)

- Claim: if $M \in \mathbb{R}^{m\times m}$ has entries $\alpha_{ij} + \beta_{ij}t$ ($\alpha_{ij}, \beta_{ij} \in \mathbb{R}$) and $\beta_{ij} \neq 0$ occurs in at most d rows, then $\det M$ is 0 or of degree $\leqslant d$.
- Induction on m. $m = 1$: $\det M = \alpha_{11} + \beta_{11}t$ (Lecture 5, Definition 7: $\det(a) = a$). No $\beta_{ij} \neq 0$ (e.g. $d = 0$): $\det M$ is a number. Otherwise expand along a row i with some $\beta_{ij} \neq 0$ (Lecture 5, Theorem 13(a): expansion along any row); each $M(i|j)$ has $\beta \neq 0$ in at most $d-1$ rows, so $\det M = \sum_j (-1)^{i+j}(\alpha_{ij} + \beta_{ij}t) \det M(i|j)$ is 0 or of degree $\leqslant d$ (induction, F1).

Induction on n; $n = 1$: $p_A = t - a_{11}$ (Lecture 5, Definition 7). For $n \geqslant 2$, as $(tI - A)(1|1) = tI_{n-1} - A(1|1)$, expanding along row 1 (Lecture 5, Theorem 13(a)) gives

$$p_A = (t - a_{11})\, p_{A(1|1)} + \sum_{j=2}^{n} (-1)^{1+j}(-a_{1j}) \det\Big((tI - A)(1|j)\Big)$$

- $j \geqslant 2$: row j of $tI - A$ loses its only t, so $(tI - A)(1|j)$ has t in at most $n-2$ rows; its determinant is 0 or of degree $\leqslant n-2$ (claim).
- Induction: $(t - a_{11})\Big(t^{n-1} - \operatorname{tr} A(1|1)\, t^{n-2} + r\Big) = t^n - \Big(a_{11} + \operatorname{tr} A(1|1)\Big)t^{n-1} + s$, where r, s are 0 or of degree $\leqslant n-3$, $\leqslant n-2$ (F1); $a_{11} + \operatorname{tr} A(1|1) = \operatorname{tr} A$.
- So $p_A = t^n - (\operatorname{tr} A)\, t^{n-1} + (0 \text{ or degree} \leqslant n-2)$.

Constant term (every n): $p_A(0) = \det(-A) = (-1)^n \det A$ (Lecture 5, Theorem 12(c)).

(b) $p_A(\lambda) = \det(\lambda I - A)$ and Theorem 2; at most n roots by (a) and F2. □

Example 4. Find the eigenvalues and eigenspaces of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & 3 \end{pmatrix}$$

Solution.

p_A, expanding along row 1 (Lecture 5, Theorem 13(a)):

$$\det(tI - A) = \det \begin{pmatrix} t-1 & -2 & -1 \\ -2 & t & 2 \\ 1 & -2 & t-3 \end{pmatrix} = (t-1)\Big(t(t-3)+4\Big) + 2\Big(-2(t-3)-2\Big) - (4-t)$$
$$= (t^3 - 4t^2 + 7t - 4) + (-4t + 8) + (t - 4) = t^3 - 4t^2 + 4t = t(t-2)^2$$

Eigenvalues 0, 2 (Theorem 3(b)).

$\lambda = 0$: reduce $A - 0I = A$; a basis of $\mathcal{N}(A)$:

$$A \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -4 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow -\frac{1}{4}R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Free unknown x_3; special solution $(1, -1, 1)'$, a basis of $\mathcal{N}(A)$ (Lecture 4, Theorem 8(c)).

$\lambda = 2$: reduce $A - 2I$; a basis of $\mathcal{N}(A - 2I)$:

$$A-2I = \begin{pmatrix} -1 & 2 & 1 \\ 2 & -2 & -2 \\ -1 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} -1 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow \frac{1}{2}R_2]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Special solution $(1, 0, 1)'$, a basis of $\mathcal{N}(A - 2I)$ (Lecture 4, Theorem 8(c)).

Definition 5. Let $A \in \mathbb{R}^{n \times n}$. By Theorem 3(a) and F3, $p_A = (t - \lambda_1) \cdots (t - \lambda_n)$ with $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, unique up to order: the **eigenvalues** of A, listed with multiplicity; they are the roots of p_A. The **algebraic multiplicity** of $c \in \mathbb{C}$ is $a_A(c) = \#\{i : \lambda_i = c\}$; c is **simple** if $a_A(c) = 1$. The real λ_i are the eigenvalues of Definition 1 (Theorem 3(b)). From now on "eigenvalue" has this meaning; eigenvectors and eigenspaces belong to real eigenvalues only.

In Example 4: eigenvalues 0, 2, 2; $a_A(0) = 1$, $a_A(2) = 2$.

Example 6. Find the eigenvalues of

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

and the eigenspace of each real eigenvalue.

Solution. Expanding along column 3 (Lecture 5, Theorem 10), and as $(t-2-i)(t-2+i) = (t-2)^2 - i^2$,

$$\det(tI - A) = \det \begin{pmatrix} t-2 & 1 & 0 \\ -1 & t-2 & 0 \\ -1 & -1 & t-1 \end{pmatrix} = (t-1)\Big((t-2)^2 + 1\Big) = (t-1)\Big(t - (2+i)\Big)\Big(t - (2-i)\Big)$$

Eigenvalues 1, $2 + i$, $2 - i$ (Definition 5); only 1 is real. For $\lambda = 1$:

$$A - I = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

So $x_1 = x_2 = 0$, x_3 free: $\mathcal{N}(A - I) = \operatorname{span}\{e_3\}$ (Lecture 2, Theorem 3(b): row operations keep the solutions); $2 \pm i$ have no eigenvectors in $\mathbb{R}^3$ (Definition 5).

3. Similar matrices

Theorem 7. Let $A, P \in \mathbb{R}^{n \times n}$, P invertible, and $B = P^{-1}AP$ (B is **similar** to A). Then $p_B = p_A$. So A, B have the same eigenvalues with the same algebraic multiplicities, $\operatorname{tr} B = \operatorname{tr} A$ and $\det B = \det A$.

Proof.

$\det(cI - B) = \det(cI - A)$ for $c \in \mathbb{R}$; $\det(FG) = \det F \det G$ (Lecture 5, Theorem 12(b)):

$$\begin{aligned} \det P^{-1} \det P = \det(P^{-1}P) = \det I = 1 \qquad & \text{(Lecture 5, Theorems 12(b), 8(d))} \\ cI - B = P^{-1}(cI)P - P^{-1}AP = P^{-1}(cI - A)P \qquad & \text{(Lecture 1, Theorem 3(a))} \\ \det(cI - B) = \det P^{-1} \det(cI - A) \det P = \det(cI - A) \qquad & \text{(Lecture 5, Theorem 12(b))} \end{aligned}$$

So $p_B = p_A$ (F2); eigenvalues, tr, det are read off p_A (Definition 5, Theorem 3(a)).

□

Theorem 8. Let $A \in \mathbb{R}^{n\times n}$ and $c \in \mathbb{R}$.

(a) If $Ax = \lambda x$, $\lambda \in \mathbb{R}$, $x \neq 0$: $A^k x = \lambda^k x$ $(k = 0, 1, 2, \ldots)$, $f(A)x = f(\lambda)x$ for every real polynomial f, and $(A + cI)x = (\lambda + c)x$.

(b) If moreover A is invertible: $\lambda \neq 0$ and $A^{-1}x = \lambda^{-1}x$.

(c) If A has the eigenvalues $\lambda_1, \ldots, \lambda_n$, then $A + cI$ has the eigenvalues $\lambda_1 + c, \ldots, \lambda_n + c$.

Proof.

(a) by induction on k, then $f(A)x$ and $(A + cI)x$:

$$\begin{aligned} A^0x = x, \qquad A^{k+1}x = A^k(Ax) = \lambda A^k x = \lambda^{k+1}x & \quad \text{(induction; Lecture 1, Theorem 3(a))} \\ f(A)x = \textstyle\sum_k c_k A^k x = \sum_k c_k \lambda^k x = f(\lambda)x & \quad (f = \textstyle\sum_k c_k t^k) \\ (A + cI)x = Ax + cx = (\lambda + c)x & \quad \text{(Lecture 1, Theorem 3(a))} \end{aligned}$$

(b) $\lambda \neq 0$, then $A^{-1}x$:

$$\begin{aligned} \lambda = 0 \implies Ax = 0,\ x \neq 0 \implies A \text{ not invertible } \text{✘} & \quad \text{((b); Lecture 2, Theorem 15)} \\ x = A^{-1}(Ax) = \lambda A^{-1}x \implies A^{-1}x = \lambda^{-1}x & \quad \text{(Lecture 1, Theorem 3(a))} \end{aligned}$$

(c) For $s \in \mathbb{R}$, with $p_A = (t - \lambda_1)\cdots(t - \lambda_n)$ (Definition 5):

$$\begin{aligned} \det\Big(sI - (A + cI)\Big) = \det\Big((s - c)I - A\Big) = p_A(s - c) & \quad \text{(Lecture 1, Definition 1)} \\ = \Big(s - (\lambda_1 + c)\Big)\cdots\Big(s - (\lambda_n + c)\Big) & \quad \text{(F4)} \end{aligned}$$

So $p_{A+cI} = \Big(t - (\lambda_1 + c)\Big)\cdots\Big(t - (\lambda_n + c)\Big)$ (F2): the eigenvalues of $A + cI$, listed with multiplicity, are $\lambda_1 + c, \ldots, \lambda_n + c$ (Definition 5, F3).

□

Theorem 9. Let $A \in \mathbb{R}^{n\times n}$ have the eigenvalues $\lambda_1, \ldots, \lambda_n$. Then $\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n$ and $\det A = \lambda_1 \cdots \lambda_n$. So A is invertible $\iff$ 0 is not an eigenvalue of A.

Proof.

Coefficients of $p_A = (t - \lambda_1)\cdots(t - \lambda_n)$ (Definition 5):

$$\begin{aligned} t^{n-1}: \quad -\operatorname{tr} A = -(\lambda_1 + \cdots + \lambda_n) & \quad \text{(Theorem 3(a), F3)} \\ t^0: \quad (-1)^n \det A = (-1)^n \lambda_1 \cdots \lambda_n & \quad \text{(Theorem 3(a), F3)} \end{aligned}$$

So A is invertible $\iff$ $\det A \neq 0$ (Lecture 5, Theorem 12(a)) $\iff$ $\lambda_i \neq 0$ for all i ($zw = 0 \Rightarrow z = 0$ or $w = 0$).

□

Example 4: $\operatorname{tr} A = 1 + 0 + 3 = 0 + 2 + 2$, $\det A = 0 \cdot 2 \cdot 2 = 0$, so A is not invertible. Example 6: $\operatorname{tr} A = 2 + 2 + 1 = 1 + (2 + i) + (2 - i)$, $\det A = 1 \cdot (2 + i)(2 - i) = 5$.

4. Independence and multiplicities

Theorem 10. Let $v_1, \ldots, v_m$ be eigenvectors of $A \in \mathbb{R}^{n\times n}$ for real eigenvalues $\lambda_1, \ldots, \lambda_m$, with $\lambda_i \neq \lambda_j$ for $i \neq j$. Then $v_1, \ldots, v_m$ are linearly independent. If A has n distinct real eigenvalues, eigenvectors for them form a basis of $\mathbb{R}^n$.

Proof. Induction on m; $m = 1$: $c_1 v_1 = 0$, $v_1 \neq 0 \Rightarrow c_1 = 0$ (Lecture 3, Proposition 2: $cv = 0 \Rightarrow c = 0$ or $v = 0$). Let $m \geqslant 2$ and $c_1 v_1 + \cdots + c_m v_m = 0$.

- A times it minus λ_m times it ($Av_i = \lambda_i v_i$; Lecture 1, Theorem 3(a)): $c_1(\lambda_1 - \lambda_m)v_1 + \cdots + c_{m-1}(\lambda_{m-1} - \lambda_m)v_{m-1} = 0$.
- Induction hypothesis: $c_i(\lambda_i - \lambda_m) = 0$, so $c_i = 0$ for $i < m$ ($\lambda_i \neq \lambda_m$).
- Then $c_m v_m = 0$ and $v_m \neq 0$, so $c_m = 0$ (Lecture 3, Proposition 2).

Last sentence: n independent vectors in $\mathbb{R}^n$ form a basis (Lecture 4, Theorem 5(d), as $\dim \mathbb{R}^n = n$ by Theorem 4). □

Definition 11. For $A \in \mathbb{R}^{n\times n}$ and $c \in \mathbb{R}$, the **geometric multiplicity** of c is $g_A(c) = \dim \mathcal{N}(A-cI) = n - \operatorname{rank}(A - cI)$ (Lecture 4, Theorem 11: $\operatorname{rank}(A - cI) + \dim \mathcal{N}(A - cI) = n$). So $g_A(c) \geqslant 1 \iff c$ is an eigenvalue (Theorem 2).

Theorem 12. If $c \in \mathbb{R}$ is an eigenvalue of $A \in \mathbb{R}^{n\times n}$, then $1 \leqslant g_A(c) \leqslant a_A(c)$. If c is simple, $g_A(c) = 1$.

Proof.
- $g = g_A(c) \geqslant 1$ (Definition 11). Extend a basis $x_1, \ldots, x_g$ of $\mathcal{N}(A - cI)$ to a basis $x_1, \ldots, x_n$ of $\mathbb{R}^n$ (Lecture 4, Theorem 5(b): an independent list extends to a basis; $\dim \mathbb{R}^n = n$, Theorem 4); $P = [x_1 \ \cdots \ x_n]$ is invertible (Lecture 4, Theorem 12(b),(d): independent columns $\iff$ rank n $\iff$ invertible).
- $Ax_j = cx_j = P(ce_j)$ for $j \leqslant g$; $Ax_j = Pm_j$ for some $m_j \in \mathbb{R}^n$ for $j > g$, as $x_1, \ldots, x_n$ span $\mathbb{R}^n$ (Lecture 1, Theorem 4(b)). So $AP = PM$ for $M = [ce_1 \ \cdots \ ce_g \ m_{g+1} \ \cdots \ m_n]$ (Lecture 1, Theorem 4(a)), and $M = P^{-1}AP$. If $g = n$: $M = cI$, $p_A = p_M = (t - c)^n$ (Theorem 7; Lecture 5, Theorem 9(b): det of a diagonal matrix is the product of its diagonal; F4, F2), and $a_A(c) = n$.
- If $g < n$, $M = \left(\begin{smallmatrix} cI_g & K \\ 0 & D \end{smallmatrix}\right)$ with $D \in \mathbb{R}^{(n-g)\times(n-g)}$, and for $s \in \mathbb{R}$, with $\det \left(\begin{smallmatrix} F & G \\ 0 & H \end{smallmatrix}\right) = \det F \det H$ (Lecture 5, Theorem 18):

$$sI - M = \begin{pmatrix} (s-c)I_g & -K \\ 0 & sI_{n-g} - D \end{pmatrix} \qquad \text{(Lecture 5, Theorem 2(a))}$$

$$\det(sI - M) = \det\Big((s-c)I_g\Big) \det(sI_{n-g} - D) = (s-c)^g p_D(s) \qquad \text{(Lecture 5, Theorems 18, 9(b))}$$

  So $p_A = p_M = (t - c)^g p_D$ (Theorem 7; F4, F2).
- With $p_D = (t - \delta_1) \cdots (t - \delta_{n-g})$ (Theorem 3(a), F3), the eigenvalues of A are c (g times) and $\delta_1, \ldots, \delta_{n-g}$ (Definition 5, F3); so $a_A(c) \geqslant g$.
- c simple: $1 \leqslant g_A(c) \leqslant a_A(c) = 1$. □

In Example 4: $g_A(2) = 1 < 2 = a_A(2)$.

5. Diagonalization

Definition 13. $A \in \mathbb{R}^{n\times n}$ is **diagonalizable** if $P^{-1}AP = \Lambda$ for an invertible $P \in \mathbb{R}^{n\times n}$ and a diagonal $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ (diagonal entries $\lambda_i \in \mathbb{R}$).

Theorem 14. Let $A \in \mathbb{R}^{n\times n}$ have the distinct real eigenvalues $c_1, \ldots, c_k$, and $W_i = \mathcal{N}(A - c_i I)$.

(a) For an invertible $P = [p_1 \ \cdots \ p_n]$ and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$: $P^{-1}AP = \Lambda \iff Ap_j = \lambda_j p_j$ ($j = 1, \ldots, n$). If so, $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A.

(b) The following are equivalent: (i) A is diagonalizable; (ii) $\mathbb{R}^n$ has a basis of eigenvectors of A; (iii) all eigenvalues of A are real and $g_A(c_i) = a_A(c_i)$ for each i; (iv) $g_A(c_1) + \cdots + g_A(c_k) = n$. In (iv), bases of $W_1, \ldots, W_k$ together form a basis of $\mathbb{R}^n$.

(c) If A has n distinct real eigenvalues, then A is diagonalizable.

Proof. (a)

$$\begin{aligned}
P^{-1}AP = \Lambda &\iff AP = P\Lambda && \text{(Lecture 1, Theorem 3(a))}\\
&\iff \operatorname{col}_j(AP) = \operatorname{col}_j(P\Lambda) \text{ for all } j && \text{(Lecture 1, Definition 1)}\\
&\iff Ap_j = \lambda_j p_j \text{ for all } j && \text{(Lecture 1, Theorem 4(a), Proposition 13(a))}\\
P^{-1}AP = \Lambda &\implies p_A = p_\Lambda = (t-\lambda_1)\cdots(t-\lambda_n) && \text{(Theorem 7; Lecture 5, Theorem 9(b); F4, F2)}
\end{aligned}$$

So $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (Definition 5).

(b)

- (i)⇔(ii): by (a); P is invertible $\iff$ its columns are independent (Lecture 4, Theorem 12(b),(d)) $\iff$ they form a basis of $\mathbb{R}^n$ (Lecture 4, Theorem 5(d)), and then they are nonzero (Lecture 3, Proposition 12(a): a list containing 0 is dependent).
- (i)⇒(iii): by (a), the eigenvalues are the real λ_j, and $a_A(c_i) = d_i = \#\{j : \lambda_j = c_i\}$. The jth entry of $(\Lambda - c_iI)x$ is $(\lambda_j - c_i)x_j$, so $\mathcal{N}(\Lambda - c_iI)$ has the basis $\{e_j : \lambda_j = c_i\}$. With $\operatorname{rank}(P^{-1}FP) = \operatorname{rank} F$ (Lecture 4, Theorem 15(d)) and $\operatorname{rank} F + \dim \mathcal{N}(F) = n$ (Lecture 4, Theorem 11), for $F \in \mathbb{R}^{n\times n}$:

$$g_A(c_i) = n - \operatorname{rank}(A - c_iI) = n - \operatorname{rank}\Big(P^{-1}(A - c_iI)P\Big) = n - \operatorname{rank}(\Lambda - c_iI) = d_i$$

- (iii)⇒(iv): each eigenvalue is one of $c_1, \dots, c_k$, so $\sum_i g_A(c_i) = \sum_i a_A(c_i) = n$.
- (iv)⇒(ii): let $w_l \in W_l$, $w_1 + \cdots + w_k = 0$. For the real polynomial $f_i = \prod_{j\neq i} \frac{t-c_j}{c_i-c_j}$ ($f_1 = 1$ if $k = 1$), $f_i(c_l) = 1$ if $l = i$ and 0 if $l \neq i$ (F4); $f(A)w_l = f(c_l)w_l$ (Theorem 8(a); trivial if $w_l = 0$). So $0 = f_i(A)(w_1 + \cdots + w_k) = f_i(c_1)w_1 + \cdots + f_i(c_k)w_k = w_i$.
- So bases $v_{l1}, \dots, v_{lg_l}$ of the W_l ($g_l = g_A(c_l)$) are together independent: $\sum_{l,r} \gamma_{lr} v_{lr} = 0$ is $w_1 + \cdots + w_k = 0$ with $w_l = \sum_r \gamma_{lr} v_{lr} \in W_l$; so all $w_l = 0$, all $\gamma_{lr} = 0$. These n eigenvectors form a basis of $\mathbb{R}^n$ (Lecture 4, Theorem 5(d)).

(c) n eigenvectors for the n eigenvalues form a basis (Theorem 10): (ii) holds. □

Not diagonalizable: Example 4 ($g_A(2) = 1 < 2 = a_A(2)$) and Example 6 ($2 \pm i$ are not real).

Example 15. Diagonalize A: find an invertible P and a diagonal Λ with $P^{-1}AP = \Lambda$, where

$$A = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix}$$

Solution.

p_A: $\text{col}_2 \leftarrow \text{col}_2 - \text{col}_3$, then take $t-2$ out of column 2 (Lecture 5, Theorem 8(a),(b), for columns by Theorem 13(b)):

$$\det(tI - A) = \det\begin{pmatrix} t-5 & 6 & 6 \\ 1 & t-4 & -2 \\ -3 & 6 & t+4 \end{pmatrix} = \det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & t-2 & -2 \\ -3 & 2-t & t+4 \end{pmatrix} = (t-2)\det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & 1 & -2 \\ -3 & -1 & t+4 \end{pmatrix}$$

$R_3 \leftarrow R_3 + R_2$ (Lecture 5, Theorem 8(b)), expand along column 2 (Lecture 5, Theorem 10):

$$\begin{aligned}
\det(tI - A) &= (t-2)\det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & 1 & -2 \\ -2 & 0 & t+2 \end{pmatrix} = (t-2)\det\begin{pmatrix} t-5 & 6 \\ -2 & t+2 \end{pmatrix}\\
&= (t-2)(t^2 - 3t + 2) = (t-1)(t-2)^2
\end{aligned}$$

Eigenvalues $1, 2, 2$; $a_A(1) = 1$, $a_A(2) = 2$ (Definition 5).

$\lambda = 1$: echelon form of $A - I$, then back substitution:

$$A - I = \begin{pmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -1 & 3 & 2 \\ 4 & -6 & -6 \\ 3 & -6 & -5 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + 3R_1]{R_2 \leftarrow R_2 + 4R_1} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 6 & 2 \\ 0 & 3 & 1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 6 & 2 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$\operatorname{rank}(A - I) = 2$ (Lecture 4, Definition 10, Remark 3: the number of nonzero rows of an echelon form, first nonzero entries not yet scaled to 1), $g_A(1) = 1$. $-x_1 + 3x_2 + 2x_3 = 0$, $3x_2 + x_3 = 0$ (Lecture 2, Theorem 3(b)); $x_3 = 3$: $\alpha_1 = (3, -1, 3)'$, a basis of $\mathcal{N}(A - I)$ (Lecture 4, Theorem 5(d)).

$\lambda = 2$: reduce $A - 2I$; special solutions:

$$A - 2I = \begin{pmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -1 & 2 & 2 \\ 3 & -6 & -6 \\ 3 & -6 & -6 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + 3R_1]{R_2 \leftarrow R_2 + 3R_1} \begin{pmatrix} -1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$g_A(2) = 3 - 1 = 2$; $\alpha_2 = (2, 1, 0)'$, $\alpha_3 = (2, 0, 1)'$, a basis of $\mathcal{N}(A - 2I)$ (Lecture 4, Theorem 8(c)).

P and Λ:

$g_A(1) + g_A(2) = 3$: $\alpha_1, \alpha_2, \alpha_3$ form a basis of $\mathbb{R}^3$ (Theorem 14(b)), so $P = [\alpha_1 \ \alpha_2 \ \alpha_3]$ is invertible (Lecture 4, Theorem 12(b),(d)), and $AP = P\Lambda$ gives $P^{-1}AP = \Lambda$ (Theorem 14(a)):

$$P = \begin{pmatrix} 3 & 2 & 2 \\ -1 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(1, 2, 2), \qquad AP = \begin{pmatrix} 3 & 4 & 4 \\ -1 & 2 & 0 \\ 3 & 0 & 2 \end{pmatrix} = P\Lambda$$

Theorem 16. If $A = P\Lambda P^{-1}$ with $P \in \mathbb{R}^{n \times n}$ invertible and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then $A^k = P\Lambda^k P^{-1} = P \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k) P^{-1}$ for $k = 0, 1, 2, \ldots$

Proof.

Induction on k; then Λ^k:

$$\begin{aligned} A^0 &= I = P\Lambda^0 P^{-1} \\ A^{k+1} &= A^k A = P\Lambda^k P^{-1} P\Lambda P^{-1} = P\Lambda^{k+1} P^{-1} && \text{(Lecture 1, Theorem 3(a))} \\ \Lambda^k &= \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k) && \text{(Lecture 1, Proposition 13(d))} \end{aligned}$$

□

Example 17. Find B^k ($k = 0, 1, 2, \ldots$) for the matrix B below. Multiplying out, $P^{-1}P = I$ (so P is invertible, Lecture 2, Theorem 15) and $BP = P\Lambda$ with $\Lambda = \operatorname{diag}(3, 3, 1, 1)$:

$$B = \begin{pmatrix} 1 & 2 & -2 & 2 \\ -2 & 3 & -2 & 0 \\ 0 & -2 & 3 & -2 \\ 2 & -2 & 2 & 1 \end{pmatrix}, \qquad P = \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 0 & -1 & 1 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

Solution. $B = BPP^{-1} = P\Lambda P^{-1}$ (Lecture 1, Theorem 3(a)).

B^k (Theorem 16; Lecture 1, Theorem 3(a)), $D = \operatorname{diag}(1, 1, 0, 0)$, $E = PDP^{-1}$:

$$\begin{aligned} B^k &= P\operatorname{diag}(3^k, 3^k, 1, 1)P^{-1} = P(3^k D + I - D)P^{-1} = 3^k E + I - E && (PP^{-1} = I) \\ B &= 3E + I - E = I + 2E, \quad \text{so } E = \tfrac{1}{2}(B - I) && (k = 1) \\ B^k &= 3^k \cdot \tfrac{1}{2}(B - I) + I - \tfrac{1}{2}(B - I) = I + \tfrac{1}{2}(3^k - 1)(B - I) && \text{(Lecture 1, Definition 1)} \end{aligned}$$

$$B^2 = I + 4(B - I) = \begin{pmatrix} 1 & 8 & -8 & 8 \\ -8 & 9 & -8 & 0 \\ 0 & -8 & 9 & -8 \\ 8 & -8 & 8 & 1 \end{pmatrix}$$

6. The products AB and BA

Theorem 18. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$ with $m \leqslant n$.

(a) $p_{BA} = t^{n-m}p_{AB}$: the eigenvalues of BA, listed with multiplicity, are those of AB and $n - m$ zeros. So AB, BA have the same nonzero eigenvalues, with the same algebraic multiplicities.

(b) If $c \neq 0$ is a real eigenvalue of AB and the columns of $X \in \mathbb{R}^{m\times g}$ form a basis of $\mathcal{N}(AB - cI_m)$, then the columns of BX are linearly independent eigenvectors of BA belonging to c. So $g_{AB}(c) = g_{BA}(c)$.

Proof. (a) For $t \in \mathbb{R}$, $t \neq 0$, and $M = \left(\begin{smallmatrix} I_n & t^{-1}B \\ A & I_m \end{smallmatrix}\right)$, with $\det I = 1$ (Lecture 5, Theorem 8(d)):

$$\begin{aligned} \det M &= \det I_n \det(I_m - AI_n^{-1}t^{-1}B) = \det(I_m - t^{-1}AB) && \text{(Lecture 5, Theorem 19(a))} \\ \det M &= \det I_m \det(I_n - t^{-1}BI_m^{-1}A) = \det(I_n - t^{-1}BA) && \text{(Lecture 5, Theorem 19(b))} \\ \det(tI_n - BA) &= t^n \det(I_n - t^{-1}BA) = t^n \det(I_m - t^{-1}AB) && \text{(Lecture 5, Theorem 12(c))} \\ &= t^{n-m}\det(tI_m - AB) && \text{(Lecture 5, Theorem 12(c))} \end{aligned}$$

So p_{BA} and $t^{n-m}p_{AB}$ have the same value at every real $t \neq 0$ (F4), and $p_{BA} = t^{n-m}p_{AB}$ (F2). With $p_{AB} = (t - \mu_1)\cdots(t - \mu_m)$ (Definition 5), $p_{BA} = (t - 0)^{n-m}(t - \mu_1)\cdots(t - \mu_m)$: the eigenvalues of BA are $\mu_1, \ldots, \mu_m$ and $n - m$ zeros (F3), and $a_{BA}(c) = a_{AB}(c)$ for $c \neq 0$ (Definition 5).

(b) Let $g = g_{AB}(c)$.

$$\begin{aligned} \operatorname{rank} X = g, \qquad ABX = cX && \text{(Lecture 4, Theorem 12(b); Lecture 1, Theorem 4(a))} \\ g = \operatorname{rank} X \geqslant \operatorname{rank}(BX) \geqslant \operatorname{rank}(ABX) && \text{(Lecture 4, Theorem 15(a))} \\ \operatorname{rank}(ABX) = \operatorname{rank}\Big(X(cI_g)\Big) = \operatorname{rank} X = g && \text{(Lecture 4, Theorem 15(d); } cI_g \text{ invertible)} \\ BA(BX) = B(ABX) = c\,BX && \text{(Lecture 1, Theorem 3(a))} \end{aligned}$$

- So the g columns of BX are independent (Lecture 4, Theorem 12(b)), lie in $\mathcal{N}(BA - cI_n)$ (Lecture 1, Theorem 4(a)) and are nonzero (Lecture 3, Proposition 12(a)): eigenvectors of BA belonging to c; $g_{BA}(c) \geqslant g$ (Lecture 4, Theorem 5(c): more than $\dim W$ vectors of W are dependent).
- The argument does not use $m \leqslant n$; with A, B exchanged it gives $g_{AB}(c) \geqslant g_{BA}(c)$. □

Example 19. Find the eigenvalues of $A'A$ and AA', and an eigenvector of AA' for each eigenvalue, where

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}, \qquad A'A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \qquad AA' = \begin{pmatrix} 2 & 1 & 1 & -1 \\ 1 & 2 & 1 & 0 \\ 1 & 1 & 2 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix}$$

Solution.

$p_{A'A}$, expanding along row 1 (Lecture 5, Theorem 13(a)), and $p_{AA'}$:

$$\det(tI_3 - A'A) = \det \begin{pmatrix} t-3 & -1 & 0 \\ -1 & t-2 & -1 \\ 0 & -1 & t-3 \end{pmatrix} = (t-3)\Big((t-2)(t-3)-1\Big) - (t-3)$$
$$= (t-3)(t^2 - 5t + 4) = (t-1)(t-3)(t-4)$$

Theorem 18(a), with A', A in place of A, B ($m = 3$, $n = 4$): $p_{AA'} = t\,p_{A'A} = t(t-1)(t-3)(t-4)$. Eigenvalues: $1, 3, 4$ for $A'A$; $0, 1, 3, 4$ for AA' (Definition 5).

$\lambda = 4, 3, 1$: eigenvectors of $A'A$, mapped by A (Theorem 18(b) with A', A in place of A, B):

$$A'A(1,1,1)' = (4,4,4)' = 4(1,1,1)', \qquad A(1,1,1)' = (2,2,2,0)'$$
$$A'A(-1,0,1)' = (-3,0,3)' = 3(-1,0,1)', \qquad A(-1,0,1)' = (-1,0,1,2)'$$
$$A'A(1,-2,1)' = (1,-2,1)' = 1(1,-2,1)', \qquad A(1,-2,1)' = (-1,2,-1,0)'$$

4, 3, 1 are simple, so each of the three vectors v is a basis of $\mathcal{N}(A'A - \lambda I_3)$ (Theorem 12; Lecture 4, Theorem 5(d)); with $X = v$: $(2,2,2,0)'$, $(-1,0,1,2)'$, $(-1,2,-1,0)'$ are eigenvectors of AA' for 4, 3, 1.

$\lambda = 0$: $\mathcal{N}(AA') = \mathcal{N}(A')$ (Lecture 4, Theorem 16(a) with A'); reduce A':

$$A' = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & -1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 2 & 2 \end{pmatrix}$$

$\operatorname{rank} A' = 3$, $\dim \mathcal{N}(A') = 4 - 3 = 1$ (Lecture 4, Definition 10, Remark 3, Theorem 11). Back substitution (Lecture 2, Theorem 3(b)), $x_4 = 1$: $x_3 = -1$, $x_2 = 0$, $x_1 = 1$; $(1,0,-1,1)'$ is a basis of $\mathcal{N}(AA')$ (Lecture 4, Theorem 5(d)).

Practice. The Exercises and Problems of Chapter 7 of the textbook.